Sahaj Bhadja

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Summary

Product-focused Full Stack Engineer with 2+ years building **enterprise AI SaaS** using **TypeScript, React, Python, LLM APIs**, owning customer-facing features and internal tools from ambiguous requirements through production.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Advanced NLP, Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Retrieval, Software Engineering, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Owned fault-tolerant asynchronous **LLM orchestration** with FastAPI workers, explicit job-state transitions, and failure isolation, processing 1K+ AI jobs at 3x throughput without cascading failures under concurrent customer usage.
- Designed multi-tenant storage with **row-level security** for 500+ users, cutting dashboard latency from 1.2s to 350ms while shipping customer-facing features through CI/CD at 99%+ reliability across production releases.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built and shipped **React and Flask product features** for uploads, workflow history, and feedback, serving 150+ instructors and 10K+ daily requests while translating user requirements into reliable interfaces.
- Created a **unified AI model layer** across OpenAI, Anthropic, and Qwen with vision fallbacks, timeout handling, usage tracking, and automated tests, reducing feedback turnaround 60% across production workflows.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built a GPT-based Decision Transformer in **PyTorch** with data pipelines over 100K+ trajectory logs, improving long-horizon planning 18% while translating research objectives into reproducible model experiments with measurable outcomes.
- Designed a multi-head **attention** encoder fusing map geometry with agent trajectories, comparing three baselines through documented evaluations and improving closed-loop stability 15% across iterative research cycles with consistent results.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built **React and Node.js SaaS workflows** for authentication, resources, billing, permissions, and notifications, synchronizing 2M+ records across locations through REST APIs and real-time WebSocket updates reliably at scale.
- Owned eight services end-to-end through deployment, adding **automated testing, monitoring, Docker, and Jenkins CI/CD** that cut release time 50% while sustaining 95%+ reliability for customer workflows in production.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built **React and TypeScript customer workflows** for booking, pricing, coupons, and payment-status updates, serving 50K+ records through Flask APIs with sub-200-millisecond responses and real-time synchronization for customers daily.
- Mapped 12+ integrations and presented **end-to-end product designs** to clients before implementation, incorporating stakeholder feedback to reduce design errors 25% while automating payment and notification workflows across delivery cycles.

Skills

Languages: TypeScript, JavaScript, Python, Java, SQL, HTML5, CSS3, C++
Frontend & Product: React.js, Next.js, TailwindCSS, Responsive Design, Product Design, User Workflows, Figma, Internal Tools
AI & ML: LLM APIs, GPT-4o, Claude, AI Agents, Prompt Engineering, RAG, Embeddings, FAISS, PyTorch, Transformers, NLP
Backend & Data: FastAPI, Flask, Node.js, REST APIs, WebSockets, PostgreSQL, MySQL, MongoDB, Redis, Authentication, Authorization
Cloud & Engineering: AWS, Azure, GCP, Docker, Kubernetes, Git, GitHub Actions, CI/CD, Jenkins, PyTest, Jest, Monitoring, Agile, Customer Collaboration

Projects

BuyerInsight Studio | *Next.js, TypeScript, FastAPI, PostgreSQL, WebSockets, LLM APIs*

- Designed a **full-stack research workspace** for creating studies, managing participant sessions, streaming interview progress, and reviewing structured findings through responsive Next.js interfaces, FastAPI services, and PostgreSQL workflows.

OpsPilot | *Python, FastAPI, LLM APIs, React, PostgreSQL, Redis*

- Designed an **AI operations agent** that converts requests into reviewable tasks, calls internal APIs, tracks approvals and execution states, and surfaces failures through asynchronous FastAPI jobs and React dashboards.

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, Clerk, PostgreSQL, GCP*

- Built an **AI application** with LLaMA-2, transformer embeddings, and FAISS semantic search over 50K+ documents with Clerk authentication, cutting retrieval latency 40% at 95% query accuracy in production.